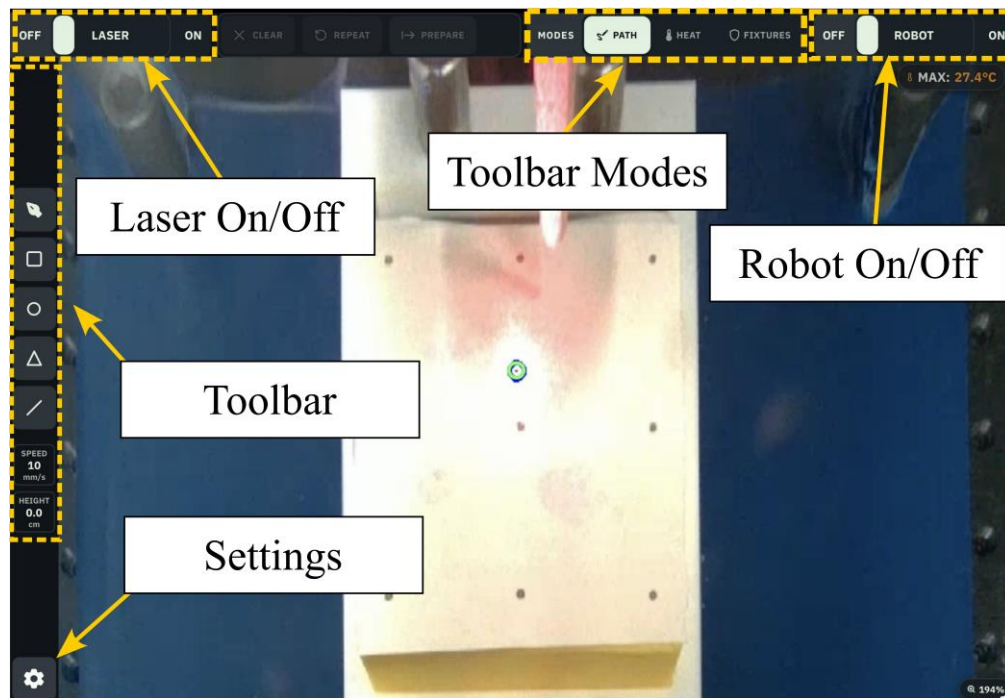


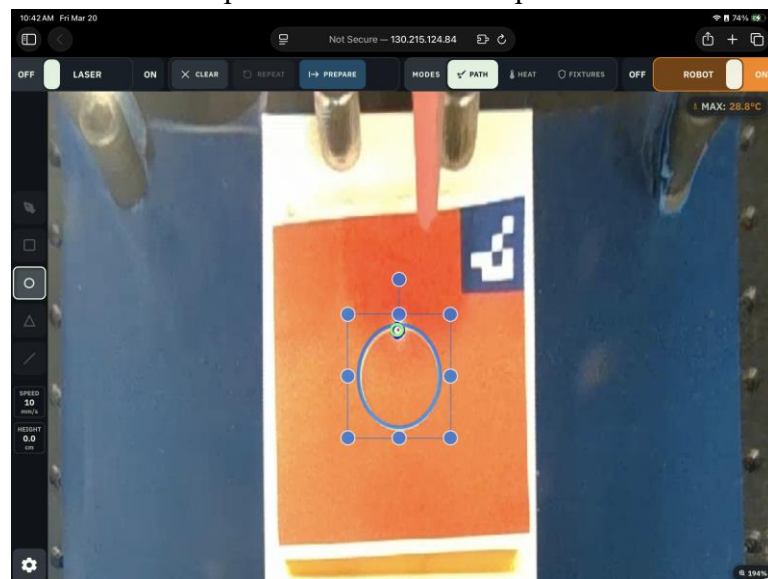
Guide on Utilization of SCALPEL-IR

Planned Path Mode

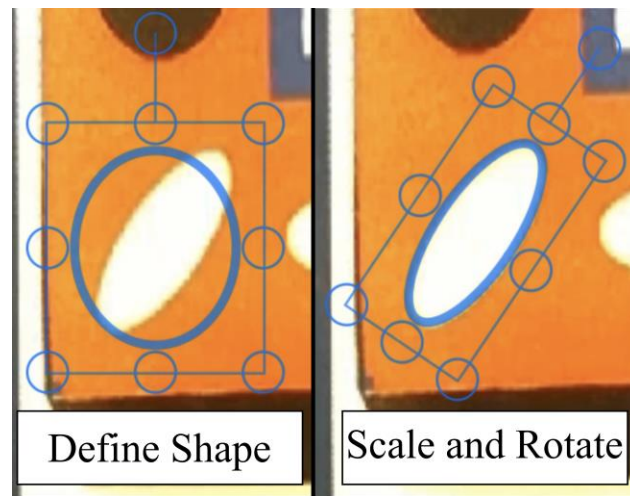
1. The following diagram shows what each button on the planned path mode does. This is the default mode that shows up when the application is launched.



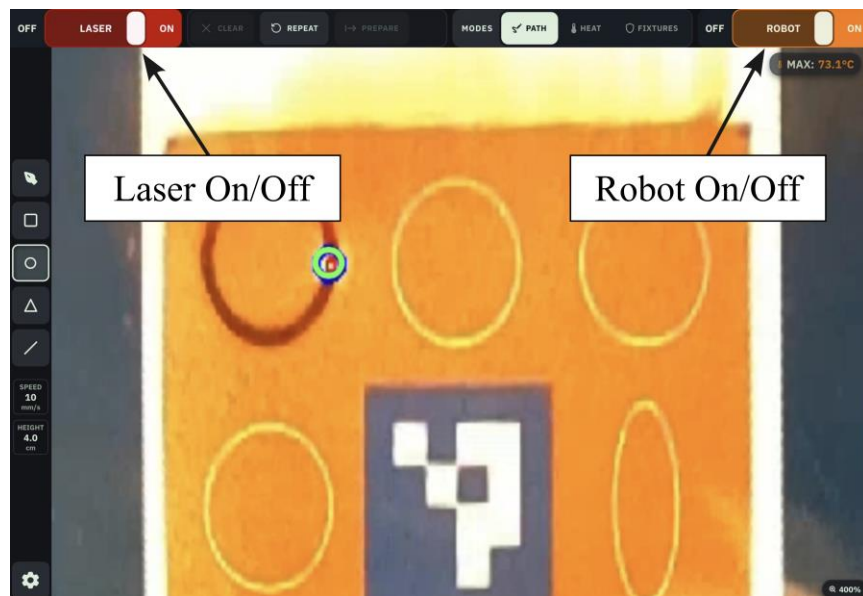
2. Use the toolbar to select a shape. Then draw that shape onto the canvas using a stylus.



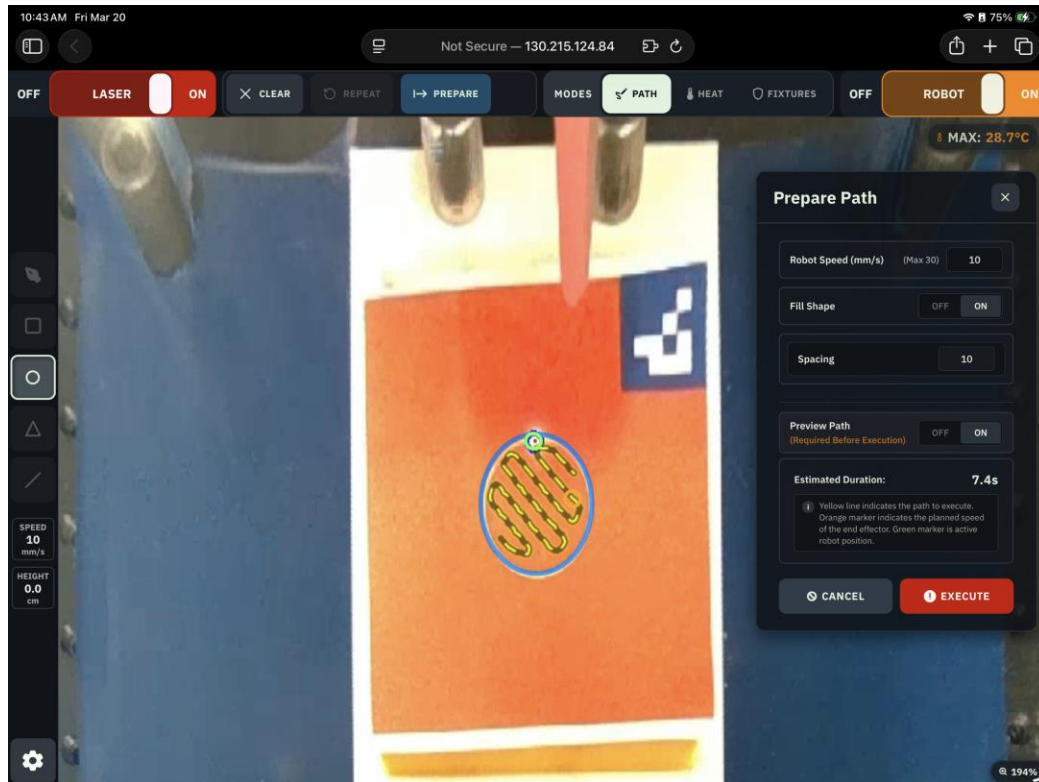
3. As seen below, shapes can be resized and rotated after creation.



4. At any time before execution turn on both the laser and the robot.



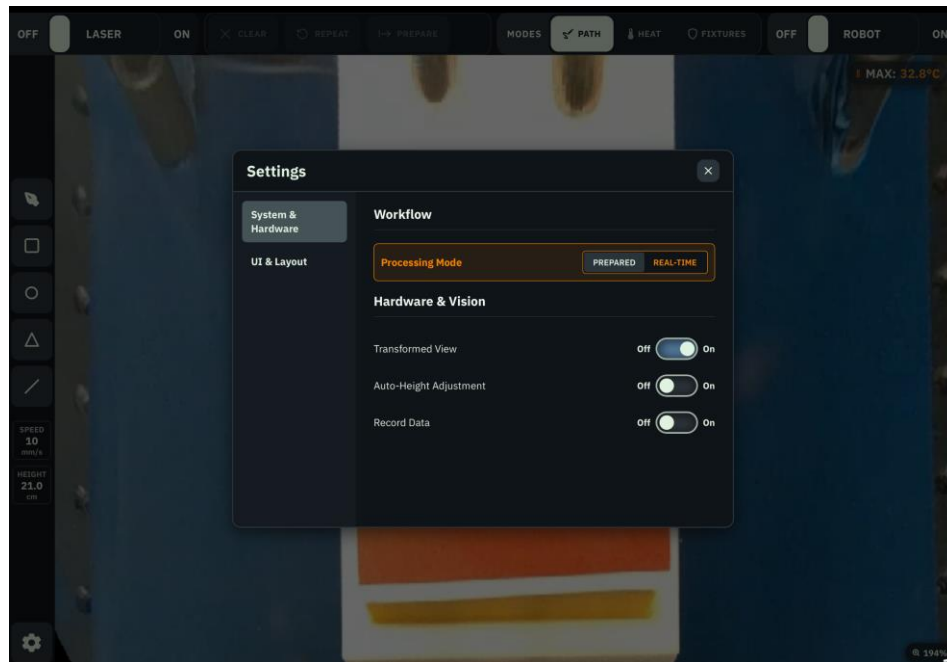
5. Click the prepare button that is on the top bar by default. This opens the prepare path window, where you can change certain parameters like speed and shape fill.



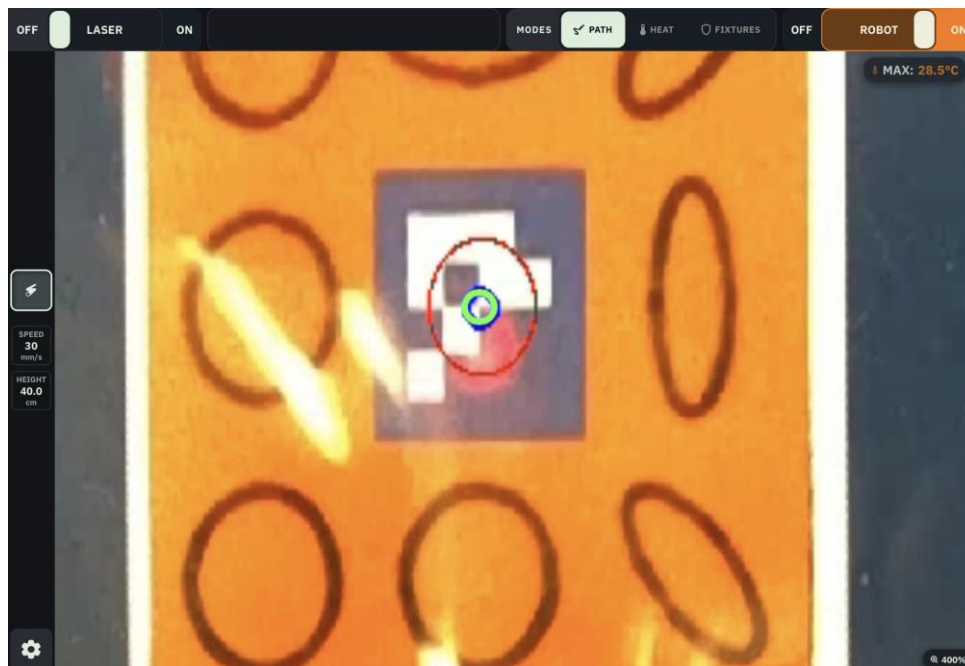
6. Before execution, click the preview path button to see what the path the laser will take
7. Click the execute button while holding down the foot pedal.

Teleoperative Mode

1. Click the settings button.
2. Change the processing mode to the real time option, which can be seen in the photo below.
3. Close out of the menu.



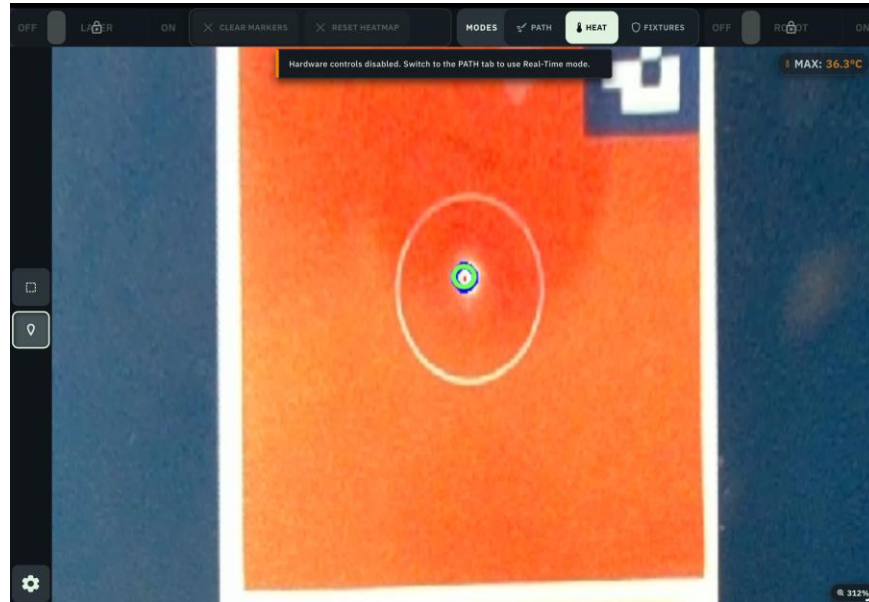
4. Now the tool bar on the left side will have a button for drawing, speed, and height
5. Before drawing, click the draw button and switch the robot on



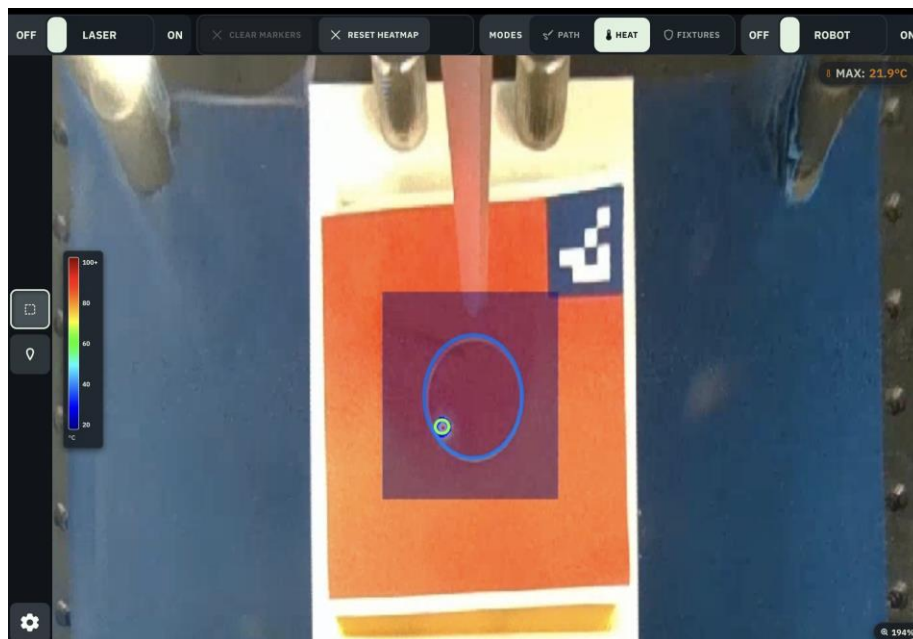
6. Once ready, click toggle the on button for the laser and hold down the foot pedal. Now when the stylus touches the screen the laser will fire where you are pointing.

Thermal Overview

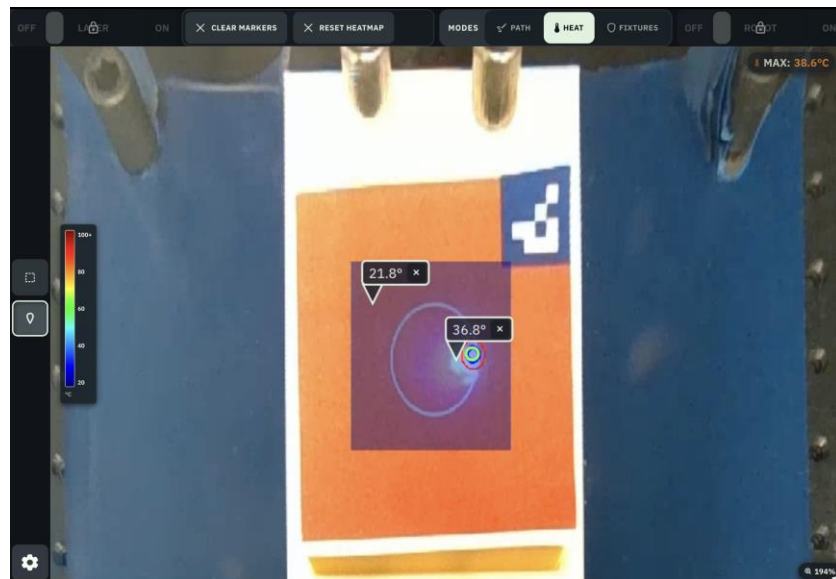
1. On either mode, click the heat button on the top toolbar



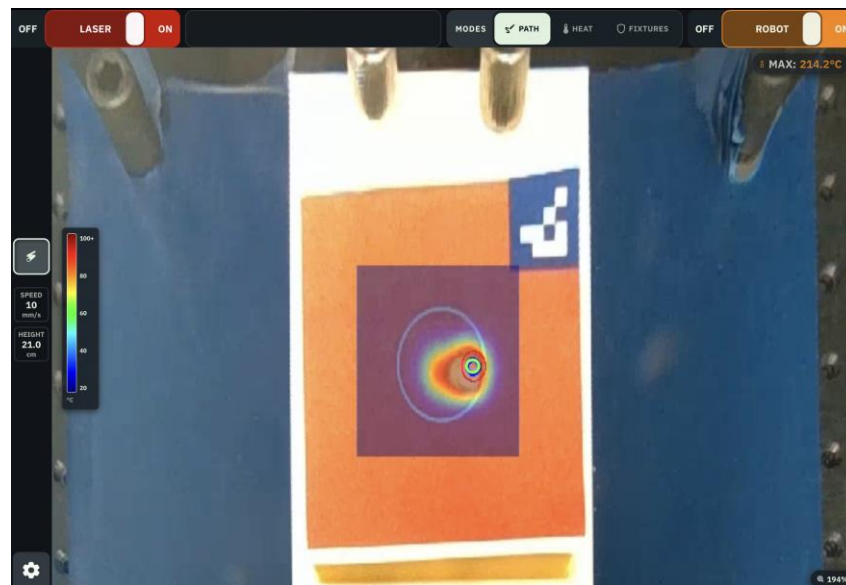
2. Now there will be two options on the left of the screen: the square one is a heat map option and the marker one is an option draw heat markers.
3. To make a heat map, click on the square icon. Then drag and drop the area that you want a heat map overlay for.



4. To make heat markers, click on the heat marker on the left and then click where you want to place the marker on the canvas. They will showcase the temperature and can be deleted with the X.

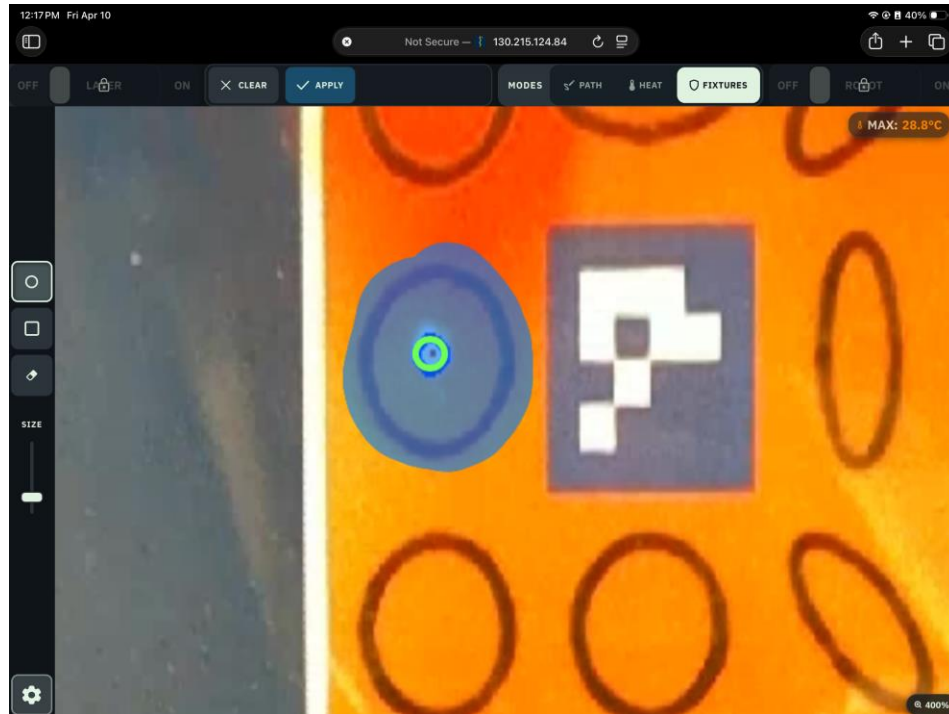


5. When ready, switch back to the drawing mode. Pictured below is an example of the heat map, while in the teleoperative mode.

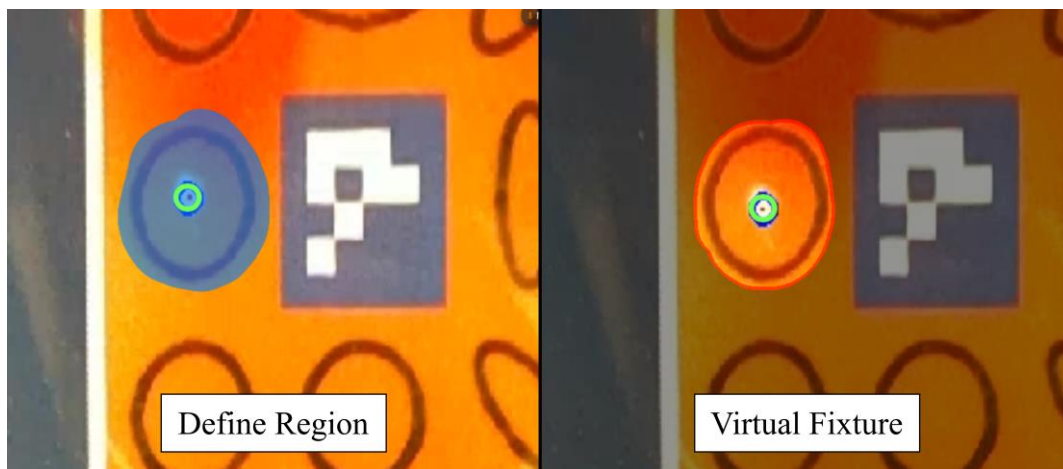


Virtual Fixtures

1. On either mode, click the fixtures tab on the top bar
2. Now on the tool bar you have the option of using a circle or square shape to draw certain regions, as can be seen below. These regions determine regions where the laser can fire.
3. If you want, the drawing shape can be increased or decreased or erased



4. When ready click the apply button
5. After clicking the applying button, the screen will highlight that section.

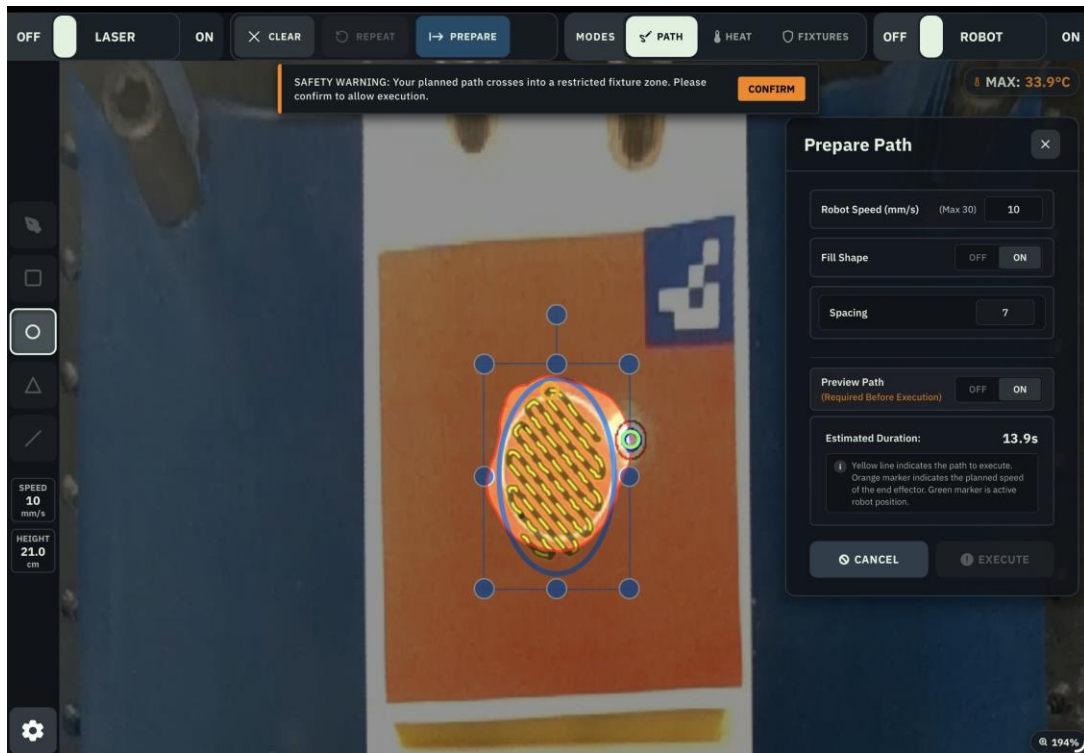


6. If the user tries to fire outside that region, the laser will automatically turn off.

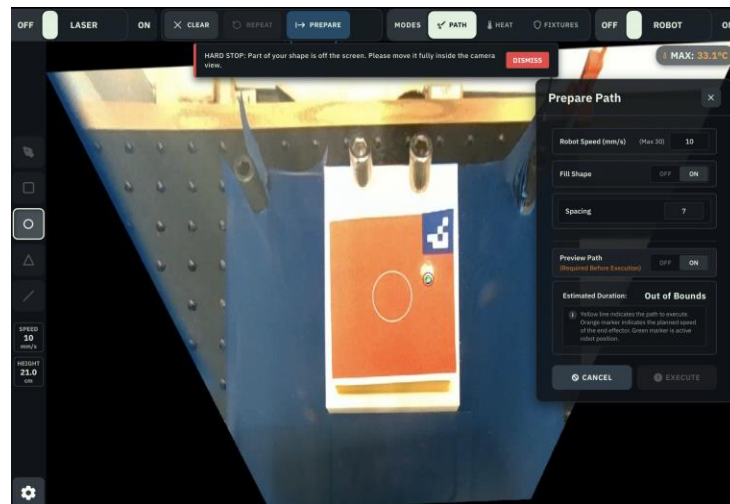
Additional Features

Safety Warnings:

1. When drawing paths outside of the virtual fixtures a warning will be given
2. If you click the confirm button, you can still fire the laser



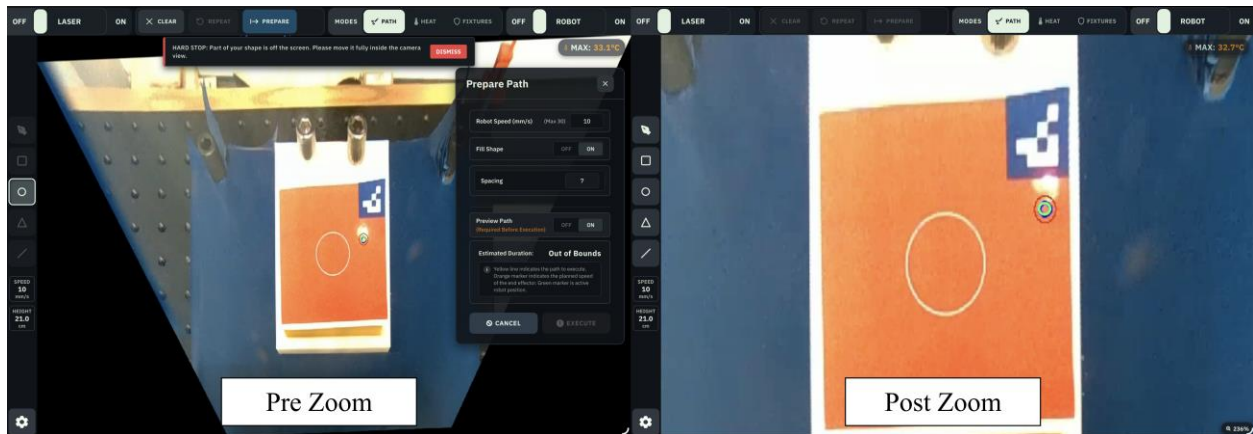
3. Additionally, if drawing outside of the canvas the following error will appear and not let you fire.



Zooming:

1. Using two fingers you can zoom in and out on the canvas

2. Pictured in the bottom right corner is the percent of zoom



UI Settings:

1. In the settings page, under the UI & Layout section, you can change the tool bars to be on the bottom and right side of the screen

